

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0313	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	A	Date:	05-10-2026

Code of Conduct

I Yoga Madha A/192425058 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Access Channels in Mobile Communication



Access Channels in Mobile Communication



1. What are Access Channels?

Access channels are the communication channels used by a mobile phone to access and communicate with a cellular network. They help the network receive requests, locate users and provide the required communication resources.

2. Types of Access Channels

RACH

Random Access Channel

Used by the mobile phone to request access to the network.

PCH

Paging Channel

Used by the network to locate a particular mobile phone.

AGCH

Access Grant Channel

Used by the network to grant access and assign communication resources.

3. How They Work

1

RACH – Request for Access

Mobile sends an access request to the network.

2

PCH – Paging

Network sends a paging message to find the mobile phone.

3

AGCH – Grant

Network grants access and assigns the required resources.

Scenario 1: Making a Call (Outgoing)



After resources are assigned, the call is established.

Scenario 2: Incoming Call (Paging)



Mobile responds and the communication is established.

Simple Example

Suppose A wants to call B.

- 1 A's mobile sends a request through **RACH**.
- 2 The network processes the request.
- 3 The network grants access through **AGCH**.
- 4 If B's mobile is to be located for an incoming call, **PCH** is used.
- 5 Finally, both mobiles can communicate.

Why are Access Channels Important?

- ✓ Connect mobile phones with the cellular network.
- ✓ Help the network handle many users.
- ✓ Help locate mobile devices.
- ✓ Allocate communication resources.
- ✓ Reduce conflicts between users.
- ✓ Help establish calls, messages and other services.

Key Points to Remember

RACH → Request

PCH → Find / Page

AGCH → Grant

*RACH requests,
PCH pages,
AGCH grants.*

Access channels act as an entry and control mechanism between mobile devices and the cellular network.